Newman-Penrose Formalism Calculations

Stephani et al., Equation (32.99a)

Metric

$$ds^2 = -\frac{4\left(\left(4\,bc^2t\sin\left(\sqrt{a}cx\right)^4 - 4\,bc^2t\sin\left(\sqrt{a}cx\right)^2 - ac^2t^{2\,c+1}\right)h\left(y\right)^2 + \left(t\sin\left(\sqrt{a}cx\right)^4 - t\sin\left(\sqrt{a}cx\right)^2\right)\left(\frac{\partial\,h}{\partial y}\right)^2\right)}{c^2t^c}\mathrm{d}u\otimes\mathrm{d}u - 2\,\cos\left(\sqrt{a}cx\right)h\left(y\right)\sin\left(\sqrt{a}cx\right)\mathrm{d}u$$
$$\otimes\,\mathrm{d}t - 2\left(2\,t\sin\left(\sqrt{a}cx\right)^2 - t\right)\sqrt{a}h\left(y\right)\mathrm{d}u\otimes\mathrm{d}x - \frac{2\,t\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial\,h}{\partial y}}{c}\mathrm{d}u\otimes\mathrm{d}y - 2\,\cos\left(\sqrt{a}cx\right)h\left(y\right)\sin\left(\sqrt{a}cx\right)\mathrm{d}t$$
$$\otimes\,\mathrm{d}u - 2\left(2\,t\sin\left(\sqrt{a}cx\right)^2 - t\right)\sqrt{a}h\left(y\right)\mathrm{d}x\otimes\mathrm{d}u + \left(t^{-c+1}\right)\mathrm{d}x\otimes\mathrm{d}x - \frac{2\,t\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial\,h}{\partial y}}{c}\mathrm{d}y\otimes\mathrm{d}u + \left(t^{c+1}\right)\mathrm{d}y\otimes\mathrm{d}y$$

Coordinates

$(x^\mu) = (u, t, x, y)$

Null Tetrad

Covariant components

$l = (1)\ du$

$$n = \left(-2\,att^c\cos\left(2\,\sqrt{a}cx\right)^2 h\left(y\right)^2 - 2\,att^c h\left(y\right)^2\sin\left(2\,\sqrt{a}cx\right)^2 - \frac{2\,bth\left(y\right)^2\sin\left(2\,\sqrt{a}cx\right)^2}{t^c} - \frac{t\sin\left(2\,\sqrt{a}cx\right)^2\frac{\partial}{\partial y}h\left(y\right)^2}{2\,c^2t^c}\right)\ du$$
$$+\left(h\left(y\right)\sin\left(2\,\sqrt{a}cx\right)\right)\ dt + \left(-2\,\sqrt{a}t\cos\left(2\,\sqrt{a}cx\right)h\left(y\right)\right)\ dx + \left(\frac{t\sin\left(2\,\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{c}\right)\ dy$$

$$m = \left(\frac{\sqrt{\frac{1}{2}}\sqrt{t}}{t^{\frac{1}{2}\,c}}\right)\ dx + \left(i\,\sqrt{\frac{1}{2}}\sqrt{t}t^{\frac{1}{2}\,c}\right)\ dy$$

$$\overline{m} = \left(\frac{\sqrt{\frac{1}{2}}\sqrt{t}}{t^{\frac{1}{2}\,c}} \right) dx + \left(-i\,\sqrt{\frac{1}{2}}\sqrt{t}t^{\frac{1}{2}\,c} \right) dy$$

Contravariant components

$$l = \left(-\frac{1}{2\,\cos\left(\sqrt{a}cx\right)h\left(y\right)\sin\left(\sqrt{a}cx\right)} \right) dt$$

$$n = \left(-1\right)\,du + \left(\frac{4\,bt\cos\left(\sqrt{a}cx\right)^{3}h\left(y\right)}{t^{c}\sin\left(\sqrt{a}cx\right)} - \frac{4\,bt\cos\left(\sqrt{a}cx\right)h\left(y\right)}{t^{c}\sin\left(\sqrt{a}cx\right)} - \frac{att^{2\,c}h\left(y\right)}{t^{c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} + \frac{t\cos\left(\sqrt{a}cx\right)^{3}\frac{\partial}{\partial y}h\left(y\right)^{2}}{c^{2}t^{c}h\left(y\right)\sin\left(\sqrt{a}cx\right)} - \frac{t\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)^{2}}{c^{2}t^{c}h\left(y\right)\sin\left(\sqrt{a}cx\right)} \right) dt$$

$$m = \left(-\frac{\sqrt{2}\sqrt{a}\sqrt{t}t^{c}\sin\left(\sqrt{a}cx\right)}{t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)} + \frac{\sqrt{2}\sqrt{a}\sqrt{t}t^{c}}{2\,t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} - \frac{i\,\sqrt{2}\sqrt{t}\frac{\partial}{\partial y}h\left(y\right)}{2\,ct^{\frac{1}{2}\,c}h\left(y\right)} \right) dt + \left(\frac{\sqrt{2}t^{\frac{1}{2}\,c}}{2\,\sqrt{t}} \right) dx + \left(\frac{i\,\sqrt{2}}{2\,\sqrt{t}t^{\frac{1}{2}\,c}} \right) dy$$

$$\overline{m} = \left(-\frac{\sqrt{2}\sqrt{a}\sqrt{t}t^{c}\sin\left(\sqrt{a}cx\right)}{t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)} + \frac{\sqrt{2}\sqrt{a}\sqrt{t}t^{c}}{2\,t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} + \frac{i\,\sqrt{2}\sqrt{t}\frac{\partial}{\partial y}h\left(y\right)}{2\,ct^{\frac{1}{2}\,c}h\left(y\right)} \right) dt + \left(\frac{\sqrt{2}t^{\frac{1}{2}\,c}}{2\,\sqrt{t}} \right) dx + \left(-\frac{i\,\sqrt{2}}{2\,\sqrt{t}t^{\frac{1}{2}\,c}} \right) dy$$

Directional Derivatives

$$D = -\frac{1}{2\,\cos\left(\sqrt{a}cx\right)h\left(y\right)\sin\left(\sqrt{a}cx\right)}\,\partial_{t}$$

$$\Delta = -\partial_{u} + \frac{4\,bt\cos\left(\sqrt{a}cx\right)^{3}h\left(y\right)}{t^{c}\sin\left(\sqrt{a}cx\right)} - \frac{4\,bt\cos\left(\sqrt{a}cx\right)h\left(y\right)}{t^{c}\sin\left(\sqrt{a}cx\right)} - \frac{att^{2\,c}h\left(y\right)}{t^{c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} + \frac{t\cos\left(\sqrt{a}cx\right)^{3}\frac{\partial}{\partial y}h\left(y\right)^{2}}{c^{2}t^{c}h\left(y\right)\sin\left(\sqrt{a}cx\right)} - \frac{t\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)^{2}}{c^{2}t^{c}h\left(y\right)\sin\left(\sqrt{a}cx\right)}\,\partial_{t}$$

$$\delta = -\frac{\sqrt{2}\sqrt{a}\sqrt{t}t^{c}\sin\left(\sqrt{a}cx\right)}{t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)} + \frac{\sqrt{2}\sqrt{a}\sqrt{t}t^{c}}{2\,t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} - \frac{i\,\sqrt{2}\sqrt{t}\frac{\partial}{\partial y}h\left(y\right)}{2\,ct^{\frac{1}{2}\,c}h\left(y\right)}\,\partial_{t} + \frac{\sqrt{2}t^{\frac{1}{2}\,c}}{2\,\sqrt{t}}\,\partial_{x} + \frac{i\,\sqrt{2}}{2\,\sqrt{t}t^{\frac{1}{2}\,c}}\,\partial_{y}$$

$$\overline{\delta} = -\frac{\sqrt{2}\sqrt{a}\sqrt{t}t^{c}\sin\left(\sqrt{a}cx\right)}{t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)} + \frac{\sqrt{2}\sqrt{a}\sqrt{t}t^{c}}{2\,t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} + \frac{i\,\sqrt{2}\sqrt{t}\frac{\partial}{\partial y}h\left(y\right)}{2\,ct^{\frac{1}{2}\,c}h\left(y\right)}\,\partial_{t} + \frac{\sqrt{2}t^{\frac{1}{2}\,c}}{2\,\sqrt{t}}\,\partial_{x} - \frac{i\,\sqrt{2}}{2\,\sqrt{t}t^{\frac{1}{2}\,c}}\,\partial_{y}$$

Spin Coefficients

$$\rho = \frac{1}{4\,t\cos\left(\sqrt{a}cx\right)h\left(y\right)\sin\left(\sqrt{a}cx\right)}$$

$$\sigma = -\frac{c}{4\,t\cos\left(\sqrt{a}cx\right)h\left(y\right)\sin\left(\sqrt{a}cx\right)}$$

$$\kappa = 0$$

$$\mu = \frac{2\,b\cos\left(\sqrt{a}cx\right)^{3}h\left(y\right)}{t^{c}\sin\left(\sqrt{a}cx\right)} - \frac{2\,b\cos\left(\sqrt{a}cx\right)h\left(y\right)}{t^{c}\sin\left(\sqrt{a}cx\right)} - \frac{at^{2\,c}h\left(y\right)}{2\,t^{c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} + \frac{\cos\left(\sqrt{a}cx\right)^{3}\frac{\partial}{\partial y}h\left(y\right)^{2}}{2\,c^{2}t^{c}h\left(y\right)\sin\left(\sqrt{a}cx\right)} - \frac{\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)^{2}}{2\,c^{2}t^{c}h\left(y\right)\sin\left(\sqrt{a}cx\right)}$$

$$\lambda = -\frac{2\,bc\cos\left(\sqrt{a}cx\right)^{3}h\left(y\right)}{t^{c}\sin\left(\sqrt{a}cx\right)} + \frac{2\,bc\cos\left(\sqrt{a}cx\right)h\left(y\right)}{t^{c}\sin\left(\sqrt{a}cx\right)} + \frac{act^{2\,c}h\left(y\right)}{2\,t^{c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} - \frac{\cos\left(\sqrt{a}cx\right)^{3}\frac{\partial}{\partial y}h\left(y\right)^{2}}{2\,ct^{c}h\left(y\right)\sin\left(\sqrt{a}cx\right)} + \frac{\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)^{2}}{2\,ct^{c}h\left(y\right)\sin\left(\sqrt{a}cx\right)}$$

$$\nu = -\frac{8i\sqrt{2}b\sqrt{t}\cos\left(\sqrt{a}cx\right)^{4}h\left(y\right)\frac{\partial}{\partial y}h\left(y\right)}{t^{\frac{3}{2}\,c}} + \frac{8i\sqrt{2}b\sqrt{t}\cos\left(\sqrt{a}cx\right)^{2}h\left(y\right)\frac{\partial}{\partial y}h\left(y\right)}{t^{\frac{3}{2}\,c}} - \frac{2i\sqrt{2}\sqrt{t}\cos\left(\sqrt{a}cx\right)^{4}\frac{\partial}{\partial y}h\left(y\right)\frac{\partial^{2}}{(\partial y)^{2}}h\left(y\right)}{c^{2}t^{\frac{3}{2}\,c}} + \frac{2i\sqrt{2}\sqrt{t}\cos\left(\sqrt{a}cx\right)^{2}\frac{\partial}{\partial y}h\left(y\right)\frac{\partial^{2}}{(\partial y)^{2}}h\left(y\right)}{c^{2}t^{\frac{3}{2}\,c}}$$

$$\alpha = -\frac{\sqrt{2}\sqrt{a}ct^{c}\cos\left(\sqrt{a}cx\right)}{4\sqrt{t}t^{\frac{1}{2}\,c}\sin\left(\sqrt{a}cx\right)} + \frac{\sqrt{2}\sqrt{a}ct^{c}\sin\left(\sqrt{a}cx\right)}{4\sqrt{t}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)} - \frac{\sqrt{2}\sqrt{a}t^{c}\cos\left(\sqrt{a}cx\right)}{4\sqrt{t}t^{\frac{1}{2}\,c}\sin\left(\sqrt{a}cx\right)} + \frac{\sqrt{2}\sqrt{a}t^{c}\sin\left(\sqrt{a}cx\right)}{4\sqrt{t}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)} + \frac{i\sqrt{2}\frac{\partial}{\partial y}h\left(y\right)}{4\sqrt{t}t^{\frac{1}{2}\,c}h\left(y\right)} - \frac{i\sqrt{2}\frac{\partial}{\partial y}h\left(y\right)}{4\,c\sqrt{t}t^{\frac{1}{2}\,c}h\left(y\right)}$$

$$\beta = 0$$

$$\tau = -\frac{\sqrt{2}\sqrt{a}ct^{c}\cos\left(\sqrt{a}cx\right)}{2\sqrt{t}t^{\frac{1}{2}\,c}\sin\left(\sqrt{a}cx\right)} - \frac{\sqrt{2}\sqrt{a}t^{c}\cos\left(\sqrt{a}cx\right)}{2\sqrt{t}t^{\frac{1}{2}\,c}\sin\left(\sqrt{a}cx\right)} + \frac{\sqrt{2}\sqrt{a}ct^{c}}{4\sqrt{t}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} - \frac{i\sqrt{2}\frac{\partial}{\partial y}h\left(y\right)}{4\sqrt{t}t^{\frac{1}{2}\,c}h\left(y\right)} + \frac{\sqrt{2}\sqrt{a}t^{c}}{4\sqrt{t}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} + \frac{i\sqrt{2}\frac{\partial}{\partial y}h\left(y\right)}{4\,c\sqrt{t}t^{\frac{1}{2}\,c}h\left(y\right)}$$

$$\pi = \frac{\sqrt{2}\sqrt{a}ct^{c}\cos\left(\sqrt{a}cx\right)}{2\sqrt{t}t^{\frac{1}{2}\,c}\sin\left(\sqrt{a}cx\right)} + \frac{\sqrt{2}\sqrt{a}t^{c}\cos\left(\sqrt{a}cx\right)}{2\sqrt{t}t^{\frac{1}{2}\,c}\sin\left(\sqrt{a}cx\right)} - \frac{\sqrt{2}\sqrt{a}ct^{c}}{4\sqrt{t}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} - \frac{i\sqrt{2}\frac{\partial}{\partial y}h\left(y\right)}{4\sqrt{t}t^{\frac{1}{2}\,c}h\left(y\right)} - \frac{\sqrt{2}\sqrt{a}t^{c}}{4\sqrt{t}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} + \frac{i\sqrt{2}\frac{\partial}{\partial y}h\left(y\right)}{4\,c\sqrt{t}t^{\frac{1}{2}\,c}h\left(y\right)}$$

$\epsilon = 0$

$$\gamma = \frac{2\,bc\cos\left(\sqrt{a}cx\right)^3 h\left(y\right)}{t^c\sin\left(\sqrt{a}cx\right)} - \frac{2\,b\cos\left(\sqrt{a}cx\right)^3 h\left(y\right)}{t^c\sin\left(\sqrt{a}cx\right)} - \frac{2\,bc\cos\left(\sqrt{a}cx\right) h\left(y\right)}{t^c\sin\left(\sqrt{a}cx\right)} + \frac{act^{2\,c}h\left(y\right)}{2\,t^c\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} + \frac{2\,b\cos\left(\sqrt{a}cx\right) h\left(y\right)}{t^c\sin\left(\sqrt{a}cx\right)} - \frac{\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)^2}{2\,ct^c h\left(y\right)} + \frac{at^{2\,c}h\left(y\right)}{2\,t^c\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} + \frac{\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)^2}{2\,c^2t^c h\left(y\right)}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{c^2}{16\,t^2\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2} + \frac{1}{16\,t^2\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2}$$

$$\Phi_{01} = -\frac{\sqrt{2}\sqrt{a}c^2t^c}{8\,t^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)} + \frac{\sqrt{2}\sqrt{a}t^c}{8\,t^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)} + \frac{\sqrt{2}\sqrt{a}c^2t^c}{16\,t^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} - \frac{i\,\sqrt{2}c\frac{\partial}{\partial y}h\left(y\right)}{16\,t^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right) h\left(y\right)^2\sin\left(\sqrt{a}cx\right)} - \frac{\sqrt{2}\sqrt{a}t^c}{16\,t^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{i\,\sqrt{2}\frac{\partial}{\partial y}h\left(y\right)}{16\,ct^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right) h\left(y\right)^2\sin\left(\sqrt{a}cx\right)}$$

$$\Phi_{02} = -\frac{ac^2t^{2\,c}\sin\left(\sqrt{a}cx\right)^2}{2\,tt^c\cos\left(\sqrt{a}cx\right)^2} + \frac{bc^2}{tt^c} - \frac{i\,\sqrt{a}c\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{2\,t\cos\left(\sqrt{a}cx\right) h\left(y\right)} - \frac{bc}{tt^c} + \frac{ac^2t^{2\,c}}{2\,tt^c\cos\left(\sqrt{a}cx\right)^2} + \frac{at^{2\,c}\sin\left(\sqrt{a}cx\right)^2}{2\,tt^c\cos\left(\sqrt{a}cx\right)^2} + \frac{i\,\sqrt{a}c\frac{\partial}{\partial y}h\left(y\right)}{4\,t\cos\left(\sqrt{a}cx\right) h\left(y\right)\sin\left(\sqrt{a}cx\right)} + \frac{i\,\sqrt{a}\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{2\,ct\cos\left(\sqrt{a}cx\right) h\left(y\right)} - \frac{at^{2\,c}}{2\,tt^c\cos\left(\sqrt{a}cx\right)^2} - \frac{ac^2t^{2\,c}}{8\,tt^c\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2} + \frac{\frac{\partial}{\partial y}h\left(y\right)^2}{8\,tt^c h\left(y\right)^2} + \frac{\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{4\,tt^c h\left(y\right)} - \frac{\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{4\,ctt^c h\left(y\right)} - \frac{i\,\sqrt{a}\frac{\partial}{\partial y}h\left(y\right)}{4\,ct\cos\left(\sqrt{a}cx\right) h\left(y\right)\sin\left(\sqrt{a}cx\right)} + \frac{at^{2\,c}}{8\,tt^c\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2} - \frac{\frac{\partial}{\partial y}h\left(y\right)^2}{8\,c^2tt^c h\left(y\right)^2}$$

$$\Phi_{10} = -\frac{\sqrt{2}\sqrt{a}c^2t^c}{8\,t^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)} + \frac{\sqrt{2}\sqrt{a}t^c}{8\,t^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)} + \frac{\sqrt{2}\sqrt{a}c^2t^c}{16\,t^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{i\,\sqrt{2}c\frac{\partial}{\partial y}h\left(y\right)}{16\,t^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right) h\left(y\right)^2\sin\left(\sqrt{a}cx\right)} - \frac{\sqrt{2}\sqrt{a}t^c}{16\,t^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} - \frac{i\,\sqrt{2}\frac{\partial}{\partial y}h\left(y\right)}{16\,ct^{\frac{3}{2}}t^{\frac{1}{2}\,c}\cos\left(\sqrt{a}cx\right) h\left(y\right)^2\sin\left(\sqrt{a}cx\right)}$$

$$
\begin{aligned}
\Phi_{11} = & -\frac{ac^2t^{2\,c}\cos\left(\sqrt{a}cx\right)^2}{4\,tt^c\sin\left(\sqrt{a}cx\right)^2} + \frac{3\,bc^2\cos\left(\sqrt{a}cx\right)^2}{4\,tt^c\sin\left(\sqrt{a}cx\right)^2} + \frac{ac^2t^{2\,c}}{4\,tt^c\sin\left(\sqrt{a}cx\right)^2} - \frac{bc\cos\left(\sqrt{a}cx\right)^2}{2\,tt^c\sin\left(\sqrt{a}cx\right)^2} + \frac{at^{2\,c}\cos\left(\sqrt{a}cx\right)^2}{4\,tt^c\sin\left(\sqrt{a}cx\right)^2} - \frac{3\,bc^2}{4\,tt^c\sin\left(\sqrt{a}cx\right)^2} - \frac{b\cos\left(\sqrt{a}cx\right)^2}{4\,tt^c\sin\left(\sqrt{a}cx\right)^2} + \frac{bc}{2\,tt^c\sin\left(\sqrt{a}cx\right)^2} \\
& - \frac{at^{2\,c}}{4\,tt^c\sin\left(\sqrt{a}cx\right)^2} - \frac{ac^2t^{2\,c}}{8\,tt^c\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2} + \frac{\cos\left(\sqrt{a}cx\right)^2\frac{\partial}{\partial y}h\left(y\right)^2}{8\,tt^ch\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2} + \frac{\cos\left(\sqrt{a}cx\right)^2\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{8\,tt^ch\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{b}{4\,tt^c\sin\left(\sqrt{a}cx\right)^2} - \frac{\cos\left(\sqrt{a}cx\right)^2\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{8\,ctt^ch\left(y\right)\sin\left(\sqrt{a}cx\right)^2} \\
& + \frac{at^{2\,c}}{8\,tt^c\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2} - \frac{\frac{\partial}{\partial y}h\left(y\right)^2}{8\,tt^ch\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2} - \frac{\cos\left(\sqrt{a}cx\right)^2\frac{\partial}{\partial y}h\left(y\right)^2}{8\,c^2tt^ch\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2} - \frac{\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{8\,tt^ch\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{8\,ctt^ch\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{\frac{\partial}{\partial y}h\left(y\right)^2}{8\,c^2tt^ch\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{12} = & -\frac{\sqrt{2}\sqrt{a}bc^2t^c\cos\left(\sqrt{a}cx\right)^4h\left(y\right)}{\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{3\sqrt{2}\sqrt{a}bc^2t^c\cos\left(\sqrt{a}cx\right)^2h\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{\sqrt{2}\sqrt{a}bt^c\cos\left(\sqrt{a}cx\right)^4h\left(y\right)}{\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{i\sqrt{2}bc\cos\left(\sqrt{a}cx\right)^3\frac{\partial}{\partial y}h\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)} - \frac{4i\sqrt{2}bc\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{\sqrt{t}t^{\frac{3}{2}\,c}} \\
& - \frac{\sqrt{2}\sqrt{a}t^c\cos\left(\sqrt{a}cx\right)^4\frac{\partial}{\partial y}h\left(y\right)^2}{4\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{2i\sqrt{2}b\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{\sqrt{t}t^{\frac{3}{2}\,c}} + \frac{\sqrt{2}a^{\frac{3}{2}}c^2t^{3\,c}h\left(y\right)}{4\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{\sqrt{2}\sqrt{a}bc^2t^ch\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{3\sqrt{2}\sqrt{a}bt^c\cos\left(\sqrt{a}cx\right)^2h\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} \\
& - \frac{i\sqrt{2}bc\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)} - \frac{i\sqrt{2}b\cos\left(\sqrt{a}cx\right)^3\frac{\partial}{\partial y}h\left(y\right)}{2\,c\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)} + \frac{3\sqrt{2}\sqrt{a}t^c\cos\left(\sqrt{a}cx\right)^2\frac{\partial}{\partial y}h\left(y\right)^2}{8\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{\sqrt{2}\sqrt{a}t^c\cos\left(\sqrt{a}cx\right)^4\frac{\partial}{\partial y}h\left(y\right)^2}{4\,c^2\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} - \frac{i\sqrt{2}act^{2\,c}\frac{\partial}{\partial y}h\left(y\right)}{8\sqrt{t}t^{\frac{3}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} \\
& + \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)^3\frac{\partial}{\partial y}h\left(y\right)^3}{8\,c\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)} - \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{c\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)} - \frac{\sqrt{2}a^{\frac{3}{2}}t^{3\,c}h\left(y\right)}{4\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{\sqrt{2}\sqrt{a}bt^ch\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{\sqrt{2}a^{\frac{3}{2}}c^2t^{3\,c}h\left(y\right)}{8\sqrt{t}t^{\frac{3}{2}\,c}\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2} \\
& + \frac{i\sqrt{2}b\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{2\,c\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)} + \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{2\,c^2\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)} - \frac{\sqrt{2}\sqrt{a}t^c\frac{\partial}{\partial y}h\left(y\right)^2}{8\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} - \frac{3\sqrt{2}\sqrt{a}t^c\cos\left(\sqrt{a}cx\right)^2\frac{\partial}{\partial y}h\left(y\right)^2}{8\,c^2\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{i\sqrt{2}at^{2\,c}\frac{\partial}{\partial y}h\left(y\right)}{8\,c\sqrt{t}t^{\frac{3}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} \\
& - \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)^3}{8\,c\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)} - \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)^3\frac{\partial}{\partial y}h\left(y\right)^3}{8\,c^3\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)} + \frac{\sqrt{2}a^{\frac{3}{2}}t^{3\,c}h\left(y\right)}{8\sqrt{t}t^{\frac{3}{2}\,c}\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2} + \frac{\sqrt{2}\sqrt{a}t^c\frac{\partial}{\partial y}h\left(y\right)^2}{8\,c^2\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)^3}{8\,c^3\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{20} = & -\frac{ac^2t^{2\,c}\cos\left(\sqrt{a}cx\right)^2}{2\,tt^c\sin\left(\sqrt{a}cx\right)^2} + \frac{bc^2}{tt^c} - \frac{i\sqrt{a}c\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{2\,th\left(y\right)\sin\left(\sqrt{a}cx\right)} - \frac{bc}{tt^c} + \frac{ac^2t^{2\,c}}{2\,tt^c\sin\left(\sqrt{a}cx\right)^2} + \frac{at^{2\,c}\cos\left(\sqrt{a}cx\right)^2}{2\,tt^c\sin\left(\sqrt{a}cx\right)^2} + \frac{i\sqrt{a}c\frac{\partial}{\partial y}h\left(y\right)}{4\,t\cos\left(\sqrt{a}cx\right)h\left(y\right)\sin\left(\sqrt{a}cx\right)} \\
& + \frac{i\sqrt{a}\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{2\,cth\left(y\right)\sin\left(\sqrt{a}cx\right)} - \frac{at^{2\,c}}{2\,tt^c\sin\left(\sqrt{a}cx\right)^2} - \frac{ac^2t^{2\,c}}{8\,tt^c\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2} - \frac{\cos\left(\sqrt{a}cx\right)^2\frac{\partial}{\partial y}h\left(y\right)^2}{8\,tt^ch\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2} + \frac{\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{4\,tt^ch\left(y\right)} - \frac{\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{4\,ctt^ch\left(y\right)} \\
& - \frac{i\sqrt{a}\frac{\partial}{\partial y}h\left(y\right)}{4\,ct\cos\left(\sqrt{a}cx\right)h\left(y\right)\sin\left(\sqrt{a}cx\right)} + \frac{at^{2\,c}}{8\,tt^c\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2} + \frac{\frac{\partial}{\partial y}h\left(y\right)^2}{8\,tt^ch\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2} + \frac{\cos\left(\sqrt{a}cx\right)^2\frac{\partial}{\partial y}h\left(y\right)^2}{8\,c^2tt^ch\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2} - \frac{\frac{\partial}{\partial y}h\left(y\right)^2}{8\,c^2tt^ch\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{21} = & -\frac{\sqrt{2}\sqrt{a}bc^2t^c\cos\left(\sqrt{a}cx\right)^4 h\left(y\right)}{\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{3\sqrt{2}\sqrt{a}bc^2t^c\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{\sqrt{2}\sqrt{a}bt^c\cos\left(\sqrt{a}cx\right)^4 h\left(y\right)}{\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{i\sqrt{2}bc\cos\left(\sqrt{a}cx\right)^3 \frac{\partial}{\partial y}h\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)} + \frac{4i\sqrt{2}bc\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{\sqrt{t}t^{\frac{3}{2}\,c}} \\
& - \frac{\sqrt{2}\sqrt{a}t^c\cos\left(\sqrt{a}cx\right)^4 \frac{\partial}{\partial y}h\left(y\right)^2}{4\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} - \frac{2i\sqrt{2}b\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{\sqrt{t}t^{\frac{3}{2}\,c}} + \frac{\sqrt{2}a^{\frac{3}{2}}c^2t^{3\,c}h\left(y\right)}{4\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{\sqrt{2}\sqrt{a}bc^2t^ch\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{3\sqrt{2}\sqrt{a}bt^c\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} \\
& + \frac{i\sqrt{2}bc\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)} + \frac{i\sqrt{2}b\cos\left(\sqrt{a}cx\right)^3 \frac{\partial}{\partial y}h\left(y\right)}{2c\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)} + \frac{3\sqrt{2}\sqrt{a}t^c\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)^2}{8\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{\sqrt{2}\sqrt{a}t^c\cos\left(\sqrt{a}cx\right)^4 \frac{\partial}{\partial y}h\left(y\right)^2}{4c^2\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{i\sqrt{2}act^{2\,c}\frac{\partial}{\partial y}h\left(y\right)}{8\sqrt{t}t^{\frac{3}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} \\
& - \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)^3 \frac{\partial}{\partial y}h\left(y\right)^3}{8c\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)} + \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{c\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)} - \frac{\sqrt{2}a^{\frac{3}{2}}t^{3\,c}h\left(y\right)}{4\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{\sqrt{2}\sqrt{a}bt^ch\left(y\right)}{2\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{\sqrt{2}a^{\frac{3}{2}}c^2t^{3\,c}h\left(y\right)}{8\sqrt{t}t^{\frac{3}{2}\,c}\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2} \\
& - \frac{i\sqrt{2}b\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{2c\sqrt{t}t^{\frac{3}{2}\,c}\sin\left(\sqrt{a}cx\right)} - \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{2c^2\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)} - \frac{\sqrt{2}\sqrt{a}t^c\frac{\partial}{\partial y}h\left(y\right)^2}{8\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} - \frac{3\sqrt{2}\sqrt{a}t^c\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)^2}{8c^2\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} - \frac{i\sqrt{2}at^{2\,c}\frac{\partial}{\partial y}h\left(y\right)}{8c\sqrt{t}t^{\frac{3}{2}\,c}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)} \\
& + \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)^3}{8c\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)} + \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)^3 \frac{\partial}{\partial y}h\left(y\right)^3}{8c^3\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)} + \frac{\sqrt{2}a^{\frac{3}{2}}t^{3\,c}h\left(y\right)}{8\sqrt{t}t^{\frac{3}{2}\,c}\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2} + \frac{\sqrt{2}\sqrt{a}t^c\frac{\partial}{\partial y}h\left(y\right)^2}{8c^2\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} - \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)^3}{8c^3\sqrt{t}t^{\frac{3}{2}\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)}
\end{aligned}
$$

$$\Phi_{22} = -\frac{4\,b^2c^2\cos\left(\sqrt{a}cx\right)^6 h\left(y\right)^2}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{8\,b^2c^2\cos\left(\sqrt{a}cx\right)^4 h\left(y\right)^2}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{4\,b^2\cos\left(\sqrt{a}cx\right)^6 h\left(y\right)^2}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{2\,abc^2\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)^2}{\sin\left(\sqrt{a}cx\right)^2} - \frac{26\,b\cos\left(\sqrt{a}cx\right)^6 \frac{\partial}{\partial y}h\left(y\right)^2}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2}$$
$$- \frac{8\,b\cos\left(\sqrt{a}cx\right)^6 h\left(y\right)\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{4\,b^2c^2\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)^2}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{8\,b^2\cos\left(\sqrt{a}cx\right)^4 h\left(y\right)^2}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{8\,b\cos\left(\sqrt{a}cx\right)^6 \frac{\partial}{\partial y}h\left(y\right)^2}{ct^{2\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{2\,abc^2h\left(y\right)^2}{\sin\left(\sqrt{a}cx\right)^2} - \frac{2\,ab\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)^2}{\sin\left(\sqrt{a}cx\right)^2}$$
$$+ \frac{52\,b\cos\left(\sqrt{a}cx\right)^4 \frac{\partial}{\partial y}h\left(y\right)^2}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{2\,b\cos\left(\sqrt{a}cx\right)^6 \frac{\partial}{\partial y}h\left(y\right)^2}{c^2t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{16\,b\cos\left(\sqrt{a}cx\right)^4 h\left(y\right)\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{4\,b^2\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)^2}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{a\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)^2}{2\,\sin\left(\sqrt{a}cx\right)^2}$$
$$- \frac{16\,b\cos\left(\sqrt{a}cx\right)^4 \frac{\partial}{\partial y}h\left(y\right)^2}{ct^{2\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{\cos\left(\sqrt{a}cx\right)^6 \frac{\partial}{\partial y}h\left(y\right)^4}{4\,c^2t^{2\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2} - \frac{4\,\cos\left(\sqrt{a}cx\right)^6 \frac{\partial}{\partial y}h\left(y\right)^2\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{c^2t^{2\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} - \frac{2\,\cos\left(\sqrt{a}cx\right)^6 \frac{\partial^2}{(\partial y)^2}h\left(y\right)^2}{c^2t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{2\,\cos\left(\sqrt{a}cx\right)^6 \frac{\partial}{\partial y}h\left(y\right)\frac{\partial^3}{(\partial y)^3}h\left(y\right)}{c^2t^{2\,c}\sin\left(\sqrt{a}cx\right)^2}$$
$$+ \frac{2\,abh\left(y\right)^2}{\sin\left(\sqrt{a}cx\right)^2} - \frac{a^2c^2t^{4\,c}h\left(y\right)^2}{4\,t^{2\,c}\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2} - \frac{26\,b\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)^2}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{4\,b\cos\left(\sqrt{a}cx\right)^4 \frac{\partial}{\partial y}h\left(y\right)^2}{c^2t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{8\,b\cos\left(\sqrt{a}cx\right)^2 h\left(y\right)\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{t^{2\,c}\sin\left(\sqrt{a}cx\right)^2}$$
$$+ \frac{2\,\cos\left(\sqrt{a}cx\right)^6 \frac{\partial}{\partial y}h\left(y\right)^2\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{c^3t^{2\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} - \frac{a\frac{\partial}{\partial y}h\left(y\right)^2}{2\,\sin\left(\sqrt{a}cx\right)^2} - \frac{a\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)^2}{2\,c^2\sin\left(\sqrt{a}cx\right)^2} + \frac{8\,b\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)^2}{ct^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{\cos\left(\sqrt{a}cx\right)^4 \frac{\partial}{\partial y}h\left(y\right)^4}{2\,c^2t^{2\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2}$$
$$+ \frac{\cos\left(\sqrt{a}cx\right)^6 \frac{\partial}{\partial y}h\left(y\right)^4}{4\,c^4t^{2\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2} + \frac{8\,\cos\left(\sqrt{a}cx\right)^4 \frac{\partial}{\partial y}h\left(y\right)^2\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{c^2t^{2\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{4\,\cos\left(\sqrt{a}cx\right)^4 \frac{\partial^2}{(\partial y)^2}h\left(y\right)^2}{c^2t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{4\,\cos\left(\sqrt{a}cx\right)^4 \frac{\partial}{\partial y}h\left(y\right)\frac{\partial^3}{(\partial y)^3}h\left(y\right)}{c^2t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{a^2t^{4\,c}h\left(y\right)^2}{4\,t^{2\,c}\cos\left(\sqrt{a}cx\right)^2\sin\left(\sqrt{a}cx\right)^2}$$
$$+ \frac{2\,b\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)^2}{c^2t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{4\,\cos\left(\sqrt{a}cx\right)^4 \frac{\partial}{\partial y}h\left(y\right)^2\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{c^3t^{2\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{a\frac{\partial}{\partial y}h\left(y\right)^2}{2\,c^2\sin\left(\sqrt{a}cx\right)^2} - \frac{\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)^4}{4\,c^2t^{2\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2} - \frac{\cos\left(\sqrt{a}cx\right)^4 \frac{\partial}{\partial y}h\left(y\right)^4}{2\,c^4t^{2\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2}$$
$$- \frac{4\,\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)^2\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{c^2t^{2\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} - \frac{2\,\cos\left(\sqrt{a}cx\right)^2 \frac{\partial^2}{(\partial y)^2}h\left(y\right)^2}{c^2t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} - \frac{2\,\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)\frac{\partial^3}{(\partial y)^3}h\left(y\right)}{c^2t^{2\,c}\sin\left(\sqrt{a}cx\right)^2} + \frac{2\,\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)^2\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{c^3t^{2\,c}h\left(y\right)\sin\left(\sqrt{a}cx\right)^2} + \frac{\cos\left(\sqrt{a}cx\right)^2 \frac{\partial}{\partial y}h\left(y\right)^4}{4\,c^4t^{2\,c}h\left(y\right)^2\sin\left(\sqrt{a}cx\right)^2}$$

$$\Lambda = \frac{3\,ac^2t^{2\,c}}{4\,tt^c} + \frac{bc^2}{4\,tt^c} + \frac{act^{2\,c}}{tt^c} - \frac{bc}{2\,tt^c} + \frac{at^{2\,c}}{4\,tt^c} + \frac{b}{4\,tt^c} - \frac{\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{8\,tt^ch\left(y\right)} + \frac{\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{8\,ctt^ch\left(y\right)}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = \frac{2i\sqrt{2}bc\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)}{\sqrt{t}t^{\frac{3}{2}c}} + \frac{i\sqrt{2}\cos\left(\sqrt{a}cx\right)\sin\left(\sqrt{a}cx\right)\frac{\partial}{\partial y}h\left(y\right)\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{2c\sqrt{t}t^{\frac{3}{2}c}h\left(y\right)}$$

$$\Psi_4 = -\frac{16b\cos\left(\sqrt{a}cx\right)^4\frac{\partial}{\partial y}h\left(y\right)^2}{t^{2c}} - \frac{8b\cos\left(\sqrt{a}cx\right)^4h\left(y\right)\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{t^{2c}} + \frac{16b\cos\left(\sqrt{a}cx\right)^2\frac{\partial}{\partial y}h\left(y\right)^2}{t^{2c}} + \frac{8b\cos\left(\sqrt{a}cx\right)^2h\left(y\right)\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{t^{2c}} - \frac{2\cos\left(\sqrt{a}cx\right)^4\frac{\partial}{\partial y}h\left(y\right)^2\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{c^2t^{2c}h\left(y\right)}$$
$$- \frac{2\cos\left(\sqrt{a}cx\right)^4\frac{\partial^2}{(\partial y)^2}h\left(y\right)^2}{c^2t^{2c}} - \frac{2\cos\left(\sqrt{a}cx\right)^4\frac{\partial}{\partial y}h\left(y\right)\frac{\partial^3}{(\partial y)^3}h\left(y\right)}{c^2t^{2c}} + \frac{2\cos\left(\sqrt{a}cx\right)^2\frac{\partial}{\partial y}h\left(y\right)^2\frac{\partial^2}{(\partial y)^2}h\left(y\right)}{c^2t^{2c}h\left(y\right)} + \frac{2\cos\left(\sqrt{a}cx\right)^2\frac{\partial^2}{(\partial y)^2}h\left(y\right)^2}{c^2t^{2c}} + \frac{2\cos\left(\sqrt{a}cx\right)^2\frac{\partial}{\partial y}h\left(y\right)\frac{\partial^3}{(\partial y)^3}h\left(y\right)}{c^2t^{2c}}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "III"